UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 001 · Introduction to partial differentiation · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a

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6. Part B · Symbol glossary

A Formula sheet

2 Limit of a function of two variables

E2.1 · Limit of f(x, y) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

GIVES	The value f approaches near (a, b).
APPLIES	The same L along every path to (a, b).
NOTE	For a polynomial (or any function continuous at the point) just substitute.

S001·9 · T001a 00:13:59

E2.2 · Two-path test (limit does not exist) ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x,y) \text{ does not exist}$$

GIVES A proof that the limit does not exist.

APPLIES Substitution gives $\frac{0}{0}$.

NOTE Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

3 Continuity of a function of two variables

E3.1 · Continuity at (a, b) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

GIVES The test for continuity.

APPLIES Three conditions: $f(a,b)$ defined, limit exists, both equal.

NOTE One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38

4 Partial derivatives: history, definition and notation

E4.1 · Partial derivatives from the definition ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h}$$

GIVES z_x (y held constant) and z_y (x held constant).

APPLIES When the limits exist.

NOTE Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

GIVES The equivalent symbols.

APPLIES $z = f(x, y)$.

NOTE The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

5 Second-order and mixed partial derivatives

E5.1 · Second-order partials ESA ★★

$$z_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}, \qquad z_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

GIVES Second derivatives in one variable.

APPLIES Differentiate z_x again in x (or z_y in y).

NOTE Same rules as ordinary second derivatives.

S001·16 · T001b 00:17:04

E5.2 · Mixed partials ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

GIVES The two crossed derivatives.

APPLIES Differentiate in one variable, then the other.

NOTE Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES The order of differentiation does not matter.

APPLIES f_{xy} and f_{yx} continuous near the point.

NOTE Called “a very important theorem” in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

B Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	x, y independent, z dependent	T1
D	Domain	Set of ordered pairs where f is defined	T1
L	Limit	Common value along every path	E2.1
(a, b)	Point of approach	Where the limit or continuity is tested	E2.1, E3.1
m	Slope of a path	Path $y = mx$ in the two-path test	E2.2, W2.2
∂	Partial ("dou")	Partial derivative sign; not δ, not d	E4.1–E5.3
h	Increment	Small change in one variable in the definition	E4.1
z_x, f_x	First partial in x	$\frac{\partial z}{\partial x}$ with y constant	E4.2
z_{xx}, z_{yy}	Second partials	$\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial y^2}$	E5.1
z_{xy}	Mixed partial	x first, then y; $= \frac{\partial^2 z}{\partial y \partial x}$	E5.2, E5.3

Remark. The slides write z_x and f_x for $\frac{\partial z}{\partial x}$; the lecture says the subscript form is the one used in problems. In the subscript notation the subscripts are read left to right in the order of differentiation; in the ∂ notation the order is read right to left (slide 001·17).